Model set -1

Basic Engineering Mathematics

Group - A

Q1.Choose the most suitable answer from the following options. **[Marks: 1 x 20 =20]**

(i) The value of $\begin{vmatrix} 1 & -1 \\ y & x \end{vmatrix}$ is

(a) x-y (b) xy (c) x+y (d) None of these

Ans.(c)

(ii) If A is a 4×5 matrix, B is a 5×4 matrix then $A\times B$ will be......... matrix.

(a) 4×4 (b) 4×5 (c) 5×4 (d) None of these

Ans.(a)

(iii) If $A=\begin{vmatrix} 7 & 9 \\ 8 & 4 \end{vmatrix}, B=\begin{vmatrix} 4 & 8 \\ -2 & 3 \end{vmatrix}$. Then the value of 2A + 5B.

(a) $\begin{vmatrix} -34 & 58 \\ -6 & 23 \end{vmatrix}$ (b) $\begin{vmatrix} 34 & 58 \\ 6 & 23 \end{vmatrix}$

(c) $\begin{vmatrix} 30 & 58 \\ 26 & 23 \end{vmatrix}$ (d) None of these

Ans.(b)

(iv) The value of cos $(\sin^{-1}x)$ is equal to

(a) $\frac{\pi}{2}-x$ (b) $\sqrt{1-x^2}$

(c) $x+\frac{\pi}{2}$ (d) None of these

Ans.(b)

(v) The value of $\sin 15^0$ is

(a) $\frac{\sqrt{3}+1}{2\sqrt{2}}$ (b) $\frac{\sqrt{3}-3}{2\sqrt{2}}$

(c) $\frac{\sqrt{3}-1}{2\sqrt{2}}$ (d) None of these

Ans.(c)

(vi) The value of cos 18^0 is equal to

(a) $-\frac{1}{4}\sqrt{10-2\sqrt{5}}$ (b) $\frac{1}{4}\sqrt{10-\sqrt{5}}$

(c) $\frac{1}{4}\sqrt{10+2\sqrt{5}}$ (d) None of these

Ans.(c)

(vii) The distance between the points (4, 2) and (3, 4) is

(a) 5 (b) $\sqrt{5}$ (c) $-5\sqrt{5}$ (d) None of these

Ans.(b)

(viii) The co-ordinate of the centre of the circle $3x^2+3y^2-8x-10y+3=0$ is.

(a) $\left(-\frac{4}{3},-\frac{5}{3}\right)$ (b) $\left(-\frac{4}{3},\frac{5}{3}\right)$

(c) $\left(\frac{4}{3},\frac{5}{3}\right)$ (d) None of these

Ans.(c)

(ix) The equation of the straight line, making intercepts 3 and 4 on the x and y axes respectively is.

(a) 4x + 3y = 12 (b) 3x + 4y = 12

(c) 3x + 4y = 1 (d) None of these

Ans.(a)

(x) The distance of the point (4, 5). From the straight the 3x-5y+7=0 is

(a) $\frac{3}{\sqrt{34}}$ (b) $\frac{6}{\sqrt{34}}$ (c) $\frac{1}{\sqrt{34}}$ (d) None of these

Ans.(b)

(xi) The differential coefficient of $\log_{10}x$.

(a) $\frac{1}{x}\log_{10}e$ (b) $\frac{1}{x}\log_e 10$

(c) $\frac{1}{x}$ (d) None of these

Ans.(a)

(xii) The value of $\lim_{x\to0}\frac{a^x-1}{x}$ is

(a) $\log_a e$ (b) $\log_e a$

(c) 1 (d) None of these

Ans.(a)

(xiii) The value of $\lim_{x\to0}\frac{\sin 3x}{2x}$ is

(a) $\frac{3}{2}$ (b) $\frac{2}{3}$

(c) 1 (d) None of these

Ans.(a)

(xiv) Differentiation of cosx w.r.t. cot x is

(a) $\cos^3 x$ (b) $\sin^3 x$

(c) $\tan^3 x$ (d) None of these

Ans.(b)

(xv) A function y = f(x) be defined in an interval (a, b) and take a point c, where a < c < b and f'(a) = 0, $f''(a)\neq0$ and $f''(c)<0$ then f(c) has

(a) maximum value

(b) minimum value

(c) neither maximum nor minimum value

(d) None of the above

Ans.(c)

(xvi)The value of sinhx is

(a) isin ix (b) – i sin ix

(c) cos ix (d) None of these

Ans.(b)

(xvii)If $\vec{a} = 2\vec{i} - 3\vec{j} - 4\vec{k}$ **and** $\vec{b} = 3\vec{i} - 3\vec{j} + \vec{k}$ **then** $\vec{a}.\vec{b} = ...$

(a) 10 (b) 11 (c) 12 (d) None of these

Ans.(b)

(xviii) Direction cosines of $\vec{a} = 5\vec{i} + 4\vec{j} + 3\vec{k}$ **is**

(a) $\left(\frac{5}{\sqrt{50}}, \frac{3}{\sqrt{50}}, \frac{4}{\sqrt{50}}\right)$ (b) $\left(\frac{5}{\sqrt{50}}, \frac{1}{\sqrt{50}}, \frac{4}{\sqrt{50}}\right)$

(c) $\left(\frac{5}{\sqrt{50}}, \frac{4}{\sqrt{50}}, \frac{3}{\sqrt{50}}\right)$ (d) None of these

Ans.(c)

(xix)The following is the data of wages in Rs. per day :5,4,7,5,8,7,5,7,9,5,7,9,10,8.The mode of the data is :

(a) 6 (b) 8 (c) 7 (d) None

Ans.(c)

(xx)The volume of a cube is increasing at the rate of 8cm³/sec.How fast the surface area increasing when the length of an edge is 12cm?

(a) $\frac{8}{3}cm^2 / \sec$ (b) $8cm^2 / \sec$

(c) $3cm^2 / \sec$ (d) None of these

Ans.(a)

Group - B

Answer all five questions : [Marks: 4 x 5 =20]

Q2.Prove that $\begin{vmatrix} a^3+1 & ab & ac \\ ac & b^1+1 & bc \\ ab & bc & c^2+1 \end{vmatrix} = 1 + a^2 + b^2 + c^2$

Ans.

$$\begin{vmatrix} a^3+1 & ab & ac \\ ac & b^1+1 & bc \\ ab & bc & c^2+1 \end{vmatrix}$$

Expand the above determiant

$$\Rightarrow (a^2+1)\{(b^2+1)(c^2+1) - bc \times bc\} - ab\{(a,b)(c^2+1) - a \times bc\}$$
$$+ ac\{ab \times bc - ac(b^2+11)\}$$

$$\Rightarrow (a^2+1)\{b^2c^2 + b^2 + c^2 + 1 - b^2c^2\} - ab\{abc^2 + ab - abc^2\}$$
$$+ ac\{ab^2c - acb^2 - ac\}$$

$$\Rightarrow (a^2+1)\{b^2+c^2+1\} - ab\{ab\} + ac(-ac)$$

$$\Rightarrow a^2b^2 + a^2c^2 + a^2 + b^2 + c^2 + 1 - a^2b^2 - a^2c^2$$

$= 1 + a^2 + b^2 + c^2$ **proved.**

OR

Q2.Find the adjoint of $\begin{vmatrix} 1 & 2 & 3 \\ 0 & 5 & 0 \\ 2 & 4 & 3 \end{vmatrix}$

Ans.Let

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 5 & 0 \\ 2 & 4 & 3 \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

$$|A| = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 5 & 0 \\ 2 & 4 & 3 \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} = \begin{bmatrix} + & - & + \\ - & + & - \\ + & - & + \end{bmatrix}$$

Then, A_{11} = Cofactor of a_{11} i.e .1st row, 1st column element,

$$= (-1)^{1+1}M_{11} \quad +\begin{vmatrix} 5 & 0 \\ 4 & 3 \end{vmatrix} = 15.$$

$$A_{12} = -\begin{vmatrix} 0 & 0 \\ 2 & 3 \end{vmatrix} = 0.$$

$$A_{13} = +\begin{vmatrix} 0 & 5 \\ 2 & 4 \end{vmatrix} = -10.$$

$$A_{21} = -\begin{vmatrix} 2 & 3 \\ 4 & 3 \end{vmatrix} = -(6-12) = 6.$$

$$A_{22} = +\begin{vmatrix} 1 & 3 \\ 2 & 3 \end{vmatrix} = (3-6) = -3.$$

$$A_{23} = -\begin{vmatrix} 1 & 2 \\ 2 & 4 \end{vmatrix} = -(4-4) = 0.$$

$$A_{31} = +\begin{vmatrix} 2 & 3 \\ 5 & 0 \end{vmatrix} = -(0-5) = -15.$$

$$A_{32} = -\begin{vmatrix} 1 & 3 \\ 0 & 0 \end{vmatrix} = 0.$$

$$A_{33} = +\begin{vmatrix} 1 & 3 \\ 0 & 0 \end{vmatrix} = 5$$

$\therefore$ The matrix of co-factor

$$= \begin{bmatrix} A_{11} & A_{12} & A_{13} \\ A_{21} & A_{22} & A_{23} \\ A_{31} & A_{32} & A_{33} \end{bmatrix} = \begin{bmatrix} 15 & 0 & -10 \\ 6 & -3 & 0 \\ -15 & 0 & 5 \end{bmatrix}$$

Hence adj A = Transpose of matrix of co factor.

$$=\begin{bmatrix} 15 & 0 & -10 \\ 6 & -3 & 0 \\ -15 & 0 & 5 \end{bmatrix}^T \begin{bmatrix} 15 & 6 & -15 \\ 0 & -3 & 0 \\ -10 & 0 & 5 \end{bmatrix}$$

Q3.If f(x) = log, $\log_e\left(\frac{1+x}{1-x}\right)$ **; Show that** $f\left(\frac{2x}{1+x^2}\right) = 2f(x)$.

Ans. We have $f(x)=\log\left(\frac{1-x}{1+x}\right)$;

$$f\left(\frac{2x}{1+x^2}\right)=\log\frac{(1-\frac{2x}{1+x^2})}{(1+\frac{2x}{1+x^2})}$$

$$=\log\frac{(1+x^2-2x)}{(1+x^2+2x)}$$

$$=\log\left(\frac{1-x}{1+x}\right)^2$$

$$=2\log\left(\frac{1-x}{1+x}\right)$$

or $$f\left(\frac{2x}{1+x^2}\right)=2f(x)$$

OR

Q3. Evaluate $\lim\limits_{x\to a}\frac{(\sqrt{a+2x}-\sqrt{3x})}{(\sqrt{3a+x}-2\sqrt{x})}$

Ans. $$\lim_{x\to a}\frac{(\sqrt{a+2x}-\sqrt{3x})}{(\sqrt{3a+x}-2\sqrt{x})}$$

$$=\lim_{x\to a}\frac{(\sqrt{a+2x}-\sqrt{3x})}{(\sqrt{3a+x}-2\sqrt{x})}\times\frac{(\sqrt{3a+x}+2\sqrt{x})}{(\sqrt{3a+x}+2\sqrt{x})}\times\frac{(\sqrt{a+2x}+\sqrt{3x})}{(\sqrt{a+2x}+\sqrt{3x})}$$

$$=\lim_{x\to a}\frac{[(a+2x)-3x]\times(\sqrt{3a+x}+2\sqrt{x})}{[(3a+x)-4x]\times(\sqrt{a+2x}+\sqrt{3x})}$$

$$=\lim_{x\to a}\frac{(a-x)\times(\sqrt{3a+x}+2\sqrt{x})}{3(a-x)\times(\sqrt{a+2x}+\sqrt{3x})}$$

$$=\lim_{x\to a}\frac{(\sqrt{3a+x}+2\sqrt{x})}{3(\sqrt{a+2x}+\sqrt{3x})}$$

$$=\frac{(\sqrt{3a+a}+2\sqrt{a})}{3(\sqrt{a+2a}+\sqrt{3a})}$$

$$=\frac{\sqrt{4a}+2\sqrt{a}}{3(\sqrt{3a}+\sqrt{3a})}=\frac{4\sqrt{a}}{3(2\sqrt{3a})}=\frac{2}{3\sqrt{3}}$$

Q4. Show that $=\lim\limits_{x\to 1}\frac{x^2-1}{|x-1|}$ **doesn't exist.**

Ans. For $x\to 1^+,(x-1)>0$ and $|x-1|=(x-1)$

For $x\to 1^-,(x-1)<0$ and $|x-1|=-(x-1)$

Now

$$\lim_{x\to 1^+}\frac{x^2-1}{|x-1|}=\lim_{x\to 1^+}\frac{x^2-1}{x-1}=\lim_{x\to 1^+}\frac{(x+1)(x-1)}{x-1}$$

$$=\lim_{x\to 1^+}(x+1)=2$$

and

$$\lim_{x\to 1^-}\frac{x^2-1}{|x-1|}=\lim_{x\to 1^-}\frac{x^2-1}{-(x-1)}=\lim_{x\to 1^+}\frac{(x+1)(x-1)}{-(x-1)}$$

$$=\lim_{x\to 1^+}[-(x+1)]=-2$$

Since $\lim\limits_{x\to 1^+}\frac{x^2-1}{|x-1|}\neq\lim\limits_{x\to 1^-}\frac{x^2-1}{|x-1|}$, limit doesn't exist.

OR

Q4. If $y=(\tan x)^{\cot x}$**, find** $\frac{dy}{dx}$.

Ans. We have $y=(\tan x)^{\cot x}$

or $\log y=\cot x\log(\tan x)$

Differentiating, we get

$$\frac{1}{y}\frac{dy}{dx}=\cot x\frac{d[\log(\tan x)]}{dx}+\log(\tan x)\frac{d[\cot x]}{dx}$$

$$=\cot x\cdot\frac{\sec^2 x}{\tan x}+\log(\tan x)(-\cos ec^2 x)$$

$$=\frac{\sec^2 x}{\tan^2 x}-\cos ec^2 x\log(\tan x)$$

$$=\frac{1}{\sin^2 x}-\cos ec^2 x\log(\tan x)$$

$$=\cos ec^2 x-\cos ec^2 x\log(\tan x)$$

or $$\frac{1}{y}\frac{dy}{dx}=\cos ec^2 x(1-\log(\tan x))$$

or $$=y\cos ec^2 x(1-\log(\tan x))$$

$$=(\tan x)^{\cot x}\cos ec^2 x(1-\log(\tan x))$$

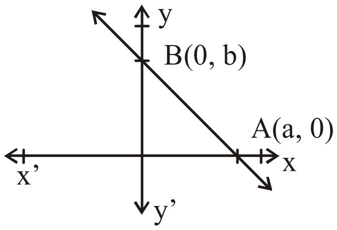
Q5.Find the equation of a straight line which cuts intercepts a and b from x and y axis respectively.

Ans. Let XOX' and YOY' be the co-ordinate axes

Let AB be the line cutting the x-axis and y-axis at A and B respectively such that $OA = a$ and $OB = b$.

Then these points are A (a, 0) and B(0, b)

So, the equation of line AB is given by



$$\frac{y-0}{x-a}=\frac{b-0}{0-a}$$

$$\Rightarrow \frac{y}{x-a}=\frac{b}{-a} \quad \Rightarrow \quad b(x-a)=-ay$$

$$\Rightarrow b(x-a)=-ay \quad \Rightarrow \quad bx+ay=ab$$

$$\Rightarrow \frac{bx}{ab}+\frac{ay}{ab}=1 \quad \text{[on dividing both sides by ab]}$$

$$\Rightarrow \boxed{\frac{x}{a}+\frac{y}{b}=1}$$

which is the required equation of line.

OR

Q5. Find the acute angle between the lines $x+3y+5=0$ and $x-2y-4=0$.

Ans. We have the equation of lines.

$x+3y+5=0$ .... (i)

$x-2y-4=0$ .... (ii)

Slope of 1st line is

$$m_1=\frac{-A}{B}=\frac{-1}{3}$$

[$\because$ slope of line $Ax+By+C=0$ is $m=-A/B$] and slope of 2nd line is

$x-2y-4=0$ is

$$m_2=\frac{1}{2}$$

If θ be the angle between two lines then.

$$\tan\theta=\left|\frac{m_1-m_2}{1+m_1m_2}\right|=\left|\frac{-\frac{1}{3}-\frac{1}{2}}{1+\left(\frac{-1}{3}\right)\left(\frac{1}{2}\right)}\right|$$

$$=\left|\frac{-5/6}{1-1/6}\right|=\left|\frac{-5/6}{5/6}\right|=|-1|$$

$$\tan\theta=1$$

$$\Rightarrow \quad \theta=\tan^{-1}(1) \quad =\tan^{-1}(\tan\pi/4)$$

$$\boxed{\theta=\pi/4}$$

Q6. Find the radius of curvature of the curve $y = 2x^3 + 3x + 7$ at the point (2, 1).

Ans. We have $y = 2x^3 + 3x + 7$ ...(1)

Differentiating wrt x we have

$$\frac{dy}{dx}=6x^2+3 \quad ...(2)$$

At (2, 1) we have

$$\left(\frac{dy}{dx}\right)_{(2,1)}=6(2)^2+3=27$$

Differentiating (2) again wrt x we have

$$\frac{d^2y}{dx^2}=12x$$

At (2, 1) we have

$$\left(\frac{d^2y}{dx^2}\right)_{(2,1)}=12\times 2=24$$

Now $$\rho=\frac{\left[1+\left(\frac{dy}{dx}\right)^2\right]^{3/2}}{\frac{d^2y}{dx^2}}=\frac{[1+(27)^2]^{3/2}}{24}$$

OR

Q6. Find median from the given frequency distribution :

Marks obtained	0-10	10-20	20-30	30-40	40-50
No. of student	8	10	22	40	20

Ans. To find the median class, we need cumulative frequencies:

Marks	0-10	10-20	20-30	30-40	40-50
Frequency	8	10	22	40	20
Cumulative frequency	8	18	40	80	100

Here $n = 100$

$\frac{n}{2}th$ i.e , 50th value lies in class 30-40, which is the median class.

$$\text{Median} = l + \frac{\frac{n}{2} - C}{f} \times c$$

$$= 30 + \frac{50-40}{40} \times 10 = 32.5$$

Group-C

Answer all five questions : [Marks: 6 x 5 =30]

Q7.Evaluate: $\lim\limits_{x\to 0} \frac{\tan x - \sin x}{x^3}$

Ans. We have $\frac{\tan x - \sin x}{x^3} = \frac{\frac{\sin x}{\cos x} - \sin x}{x^3}$

$$= \frac{\sin x(1-\cos x)}{x^3 \cos x}$$

$$= \frac{\sin x(2\sin^2 \frac{x}{2})}{x^3 \cos x}$$

$$= \frac{\sin x}{x} \cdot \frac{2\sin^2 \frac{x}{2}}{x^2} \cdot \frac{1}{\cos x}$$

$$= \frac{\sin x}{x} \cdot \frac{2\sin^2 \frac{x}{2}}{(\frac{x}{2})^2} \cdot \frac{1}{4} \cdot \frac{1}{\cos x}$$

$$= \frac{1}{2} \cdot \frac{\sin x}{x} \cdot \frac{2\sin^2 \frac{x}{2}}{(\frac{x}{2})^2} \cdot \frac{1}{\cos x}$$

$$\lim_{x\to 0} \frac{\tan x - \sin x}{x^3} = \lim_{x\to 0} \frac{1}{2} \cdot \frac{\sin x}{x} \cdot \frac{2\sin^2 \frac{x}{2}}{(\frac{x}{2})^2} \cdot \frac{1}{\cos x}$$

$$= \frac{1}{2}\left(\lim_{x\to 0} \frac{\sin x}{x}\right)\left(\lim_{x\to 0} \frac{\sin^2 \frac{x}{2}}{(\frac{x}{2})^2}\right)\left(\lim_{x\to 0} \frac{1}{\cos x}\right)$$

$$= \left(\frac{1}{2} \times 1 \times 1^2 \times 1\right) = \frac{1}{2}$$

OR

Q7.Find derivative of f(x) =sin^{-1} x using the first principle of differentiation.

Ans. Let $y = \sin^{-1} x,\ x \in [-1, 1],\ y \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

$x = \sin y,$

Now, by definition

$$\frac{dx}{dy} = \lim_{h\to 0} \frac{f(y+h) - f(y)}{h}$$

$$= \lim_{h\to 0} \frac{\sin(y+h) - \sin y}{h}$$

$$= \lim_{h\to 0} \frac{2\cos \frac{y+h+y}{2} \sin \frac{y+h-y}{2}}{h}$$

$$= \lim_{h\to 0} \cos\left(y + \frac{h}{2}\right) \lim_{\frac{h}{2}\to 0} \frac{\sin \frac{h}{2}}{\frac{h}{2}}$$

$$= \cos(y+0) \cdot 1 = \cos y$$

Also $\cos y = \pm\sqrt{1 - \sin^2 y}$ $\quad [\cos^2 y + \sin^2 y = 1]$

But $y \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] \Rightarrow \cos y \ge 0$

$$\cos y = \sqrt{1 - \sin^2 y} = \sqrt{1 - x^2}$$

We know that $\frac{dy}{dx} = \frac{1}{\frac{dx}{dy}}$, provided $\frac{dx}{dy} \ne 0$ i.e.

$\cos y \ne 0$ i.e. $y \ne \pm\frac{\pi}{2}$ i.e. $x \ne \pm 1$ (since $x = \sin y$)

Hence $\frac{dy}{dx} = \frac{1}{\frac{dx}{dy}} = \frac{1}{\cos y}$

$$= \frac{1}{\sqrt{1-x^2}},\ x \in [-1, 1] \text{ and } x \ne \pm 1$$

Thus,

$$\frac{d}{dx}\left(\sin^{-1} x\right) = \frac{1}{\sqrt{1-x^2}},\ x \in (-1, 1) \text{ i.e. } |x| < 1$$

Q8.If the following function f(x) is continuous at x =0, find the value of k.

$$f(x) = \begin{cases} \frac{1-\cos 2x}{2x^2}, & x \ne 0 \\ k & , x = 0 \end{cases}$$

Ans. Given that function is continuous at $x = 0$

f(x) is continuous at $x = 0$

i.e. $\lim\limits_{x\to 0} f(x) = f(0) = K$

Limit at $x \to 0$

$$\lim_{x\to 0} f(x) = \lim_{x\to 0} \frac{(1-\cos 2x)}{2x^2}$$

$$= \lim_{x\to 0} \frac{2\sin^2 x}{2x^2} \quad (\ldots(1-\cos 2x) = 2\sin^2 x)$$

$$= \lim_{x\to 0}\left(\frac{\sin x}{x}\right)^2$$

$$=1$$

$$= \lim_{x\to 0} f(x) = f(0) = k = 1$$

$\therefore \quad K = 1$

OR

Q8.If y =(x)$^{\cos x}$ + (sinx)$^{\tan x}$, Find $\frac{dy}{dx}$.

Ans. We have $y = x^{\cos x} + (\sin x)^{\tan x}$

Let $y = u + v$

where $u = x^{\cos x}$ and $v = (\sin x)^{\tan x}$

Now $u = x^{\cos x}$

$$\log u = \cos x \log x$$

$$\frac{1}{u}\frac{du}{dx} = \frac{\cos x}{x} + \log x(-\sin x)$$

or $$\frac{du}{dx} = u\left(\frac{\cos x}{x} - \sin x \log x\right)$$

or $$\frac{du}{dx} = x^{\cos x}\left(\frac{\cos x}{x} - \sin x \log x\right)$$

$$\log v = \tan \log \sin x$$

Differentiate wrt x we have

$$\frac{1}{v}\frac{dv}{dx} = \tan x \cdot \frac{\cos x}{\sin x} + \log \sin x \cdot \sec^2 x$$

or $$\frac{dv}{dx} = v\left(\tan x \cot x + \sec^2 x \log \sin x\right)$$

$$= (\sin x)^{\tan x}(1 + \sec^2 x \log \sin x)$$

Thus $$\frac{dy}{dx} = \frac{du}{dx} + \frac{dv}{dx}$$

$$= x^{\cos x}\left(\frac{\cos x}{x} - \sin x \log x\right)$$

$$+ (\sin x)^{\tan x}(1 + \sec^2 x \log \sin x)$$

Q9.The surface area of a spherical bubble is increasing at 2cm³/sec.At what rate is the volume of the bubble increasing when the radius of the bubble is 6cm?

Ans. Let r be the radius, S the surface area and V the volume of the spherical bubble at any time t.

Surface area, $S = 4\pi r^2$

$$\frac{dS}{dt} = 4\pi \times 2r \times \frac{dr}{dt} = 8\pi r\frac{dr}{dt}$$

$$\frac{dr}{dt} = \frac{1}{8\pi r}\frac{dS}{dt}$$

As $\frac{dS}{dt} = 2cm^2 / sec$

Thus $\frac{dr}{dt} = \frac{1}{8\pi r} \times 2 = \frac{1}{4\pi r}$

Volume $V = \frac{4}{3}\pi r^3$

$$\frac{dV}{dt} = \frac{4\pi}{3} \times 3r^2 \times \frac{dr}{dt}$$

$$= 4\pi r^2 \times \frac{1}{4\pi r}$$

$$= r\ cm^3 / sec$$

Hence $\left(\frac{dV}{dt}\right)_{r=6cm} = 6cm^3 / sec$

Hence, the rate at which the volume of the bubble is increasing at the instant its radius is 6 cm³/sec.

OR

Q9.Prove that the maximum value of $\left(\frac{1}{x}\right)^x$ is $e^{\frac{1}{e}}$.

Ans. Let $y = \left(\frac{1}{x}\right)^x$

Taking log on both side

$$\log y = x \log\left(\frac{1}{x}\right) = -x \log e^x \qquad ...(1)$$

Differentiating (1) w.r.t x.

$$\frac{1}{y}\frac{dy}{dx} = -\left[x \cdot \frac{1}{x} + \log e^x\right] = -(1 + \log e^x) \qquad ...(2)$$

$$\Rightarrow \frac{dy}{dx} = -\left(\frac{1}{x}\right)^x (1 + \log e^x)$$

Again differentiating (2) w.r.t x.

$$\frac{1}{y}\frac{d^2y}{dx^2} - \frac{1}{y^2}\left(\frac{dy}{dx}\right)^2 = -\frac{1}{x} \qquad ...(3)$$

from eq^n (2), we get

$$\frac{dy}{dx} = -y(1 + \log e^x) = -\left(\frac{1}{x}\right)^x (1 + \log e^x)$$

For the maximum or minimum values of y,

$$\frac{dy}{dx}=0$$

Therefore, $\left(\frac{1}{x}\right)^{x}(1+\log e^{x})=0$

However B $(1/x)^{x}\neq 0$ for any value of x. Therefore

$$1+\log e^{x}=0$$

$$\log e^{x}=-1$$

$$x=e^{-1}=\frac{1}{e}$$

when $x=1/e$ in eq^{n} (3)

$$\frac{1}{y}\frac{d^2y}{dx^2}-0=-e$$

Therefore, $\frac{d^2y}{dx^2}=-e(e)^{1/e}$

Hence, y is max^{m} when

$x=1/e$ and the max^{m}

Value of $y-e^{1/e}$

Q10.Solve the following system of equations by matrix method.

$$x+2y-3z=-4$$

$$2x+3y+2z=2$$

$$3x-3y-4z=11$$

Ans. The given equation can be written as

$$AX=B$$

$$X=A^{-1}.B$$

Where,

$$A=\begin{vmatrix}1&2&-3\\2&3&2\\3&-3&-4\end{vmatrix};\ X=\begin{vmatrix}x\\y\\z\end{vmatrix}\ B=\begin{vmatrix}-4\\2\\11\end{vmatrix}$$

Determinant of A:

$$|A|=\begin{vmatrix}1&2&-3\\2&3&2\\3&-3&-4\end{vmatrix}$$

$= 1(-12+6) -2(-8-6) -3(-6-9)$

$= -6+28+45$

$= 22+45 = 67\neq 0$

Cofactors of A :

Co-factor of 1st row :

$A_{11}=(-1)^{2}(-12+6)=-6$

$A_{12}=(-1)^{1+2}(-8-6)=14$

$A_{13}=(-1)^{1+3}(-6-9)=-15$

Co-factor of 2nd row :

$A_{21}=(-1)^{2+1}(-8-9)=17$

$A_{22}=(-1)^{2+2}(-4+9)=5$

$A_{23}=(-1)^{2+3}(-3-6)=9$

Co-factor of 3rd row :

$A_{31}=(-1)^{3+1}(4+9)=13$

$A_{32}=(-1)^{3+2}(2+6)=-8$

$A_{33}=(-1)^{3+3}(3-4)=-1$

$\therefore$ Co-factor matrix

$$C=\begin{bmatrix}-6&14&-15\\17&5&9\\13&-8&-1\end{bmatrix}=\begin{bmatrix}-6&17&13\\14&5&-8\\-15&9&-1\end{bmatrix}$$

$$A^{-1}=\frac{adj.\,of\,A}{|A|}=\frac{1}{67}\begin{bmatrix}-6&17&13\\14&5&-8\\-15&9&-1\end{bmatrix}$$

$$X=\begin{bmatrix}x\\y\\z\end{bmatrix}=A^{-1}.B=\frac{1}{67}\begin{bmatrix}-6&17&13\\14&5&-8\\-15&9&-1\end{bmatrix}\begin{bmatrix}-4\\2\\11\end{bmatrix}$$

$$=\frac{1}{67}\begin{bmatrix}24+34+143\\-56+10-88\\+60+18-11\end{bmatrix}=\frac{1}{67}\begin{bmatrix}201\\-134\\+67\end{bmatrix}$$

$$x=\frac{201}{67};\ y=\frac{-134}{67};\ z=\frac{+67}{67}$$

$$x=3;\ y=-2;\ z=1\ \left.\begin{matrix}x=3\\y=-2\\z-1\end{matrix}\right\}$$

OR

Q10.Find the equation of straight line which passes through the point of intersection of the lines $2x+3y-13=0,\ 5x-y-7=0$ and perpendicular to the line $2x-5y+7=0$.

Ans. We have

$2x+3y-13=0$ ....(i)

$5x-y-7=0$ ...(ii)

For point of intersection of (i) and (ii)

$2x+3y=13$ .....(i) × 5

$5x-y=7$ ....(ii) × 2

$$10x+15y=65$$

$$10x-2y=14$$

$$\underline{-\quad+\quad-}$$

$$17y=51\qquad\Rightarrow\ y=3$$

Putting y = 3 in (i) and get

$$2x+3\times 3=13$$
$$2x=13-9$$
$$x=2$$

$\therefore$ Point of intersection of (i) and (ii) is (2, 3)

Also given lines is $2x-5y+7=0$

$$m_1 = slope = \frac{-A}{B} = \frac{-2}{-5} = \frac{2}{5}$$

and slope of the line $\perp r$ to the give line is

$$m_2 = \frac{-1}{m_1} \quad [\because m_1.m_2=-1]$$

Now using slope point from equation of line passing through the point (2, 3) and slope –5/2 is

$$y-y_1=m(x-x_1)$$
$$\Rightarrow \quad y-3=\frac{-5}{2}(x-2)$$
$$\Rightarrow \quad 2y-6=-5x+10$$
$$\Rightarrow \quad 5x+2y-16=0$$

which is required equation of line.

Q11.Find the equation of a circle passes through the centre of the circle $x^2 + y^2 + 8x + 10y - 7 = 0$ and concentric with the circle $x^2 + y^2 - 4x - 6y = 0$

Ans. Equations of given circles are

$$x^2+y^2+8x+10y-7=0 \quad ....(i)$$
$$x^2+y^2-4x-6y=0 \quad ...(ii)$$

Comparing (i) with general equation of motion and get, $2g=8$ $\quad 2f=10$

$g=4$ $\quad f=5$

$\therefore$ Centre of (i) is $=(-g,-f)=(-4,-5)$

Also the required circle is concentric with the circle (ii) and passing through the centre of (i) i.e., (–4, –5).

Now comparing (ii) with $x^2+y^2+2gx+2fy+c=0$ and get

$$2g=-4 \qquad 2f=-6$$
$$g=-2 \qquad f=-3$$

So, the centre of required circle is (2, 3)

Which is passes through the pt (–4, –5)

So, radius of required circle is

$$r=\sqrt{(-4-2)^2+(-5-3)^2}$$
$$=\sqrt{36+64}=\sqrt{100}=10\,\text{unit}$$

Now using centre radius form equation of circle is

$$(x-h)^2+(y-k)^2=r^2$$
$$\therefore \quad (x-2)^2+(y-3)^2=100$$
$$\Rightarrow x^2+y^2-4x-6y+13-100=0$$
$$\Rightarrow \boxed{x^2+y^2-4x-6y-87=0}$$

Which is required equation of circle.

OR

Q11.(a) The mean of the following distribution is 26.Find the value of p and also Find the mode of

X_i	0	1	2	3	4	5
F_i	3	3	P	7	P-1	4

Ans. We know that

$$\text{Mean} = \frac{\text{Sum of observation}}{\text{Number of observation}}$$

$$26=\frac{3+3+P+7+(P-1)+4}{6}$$

$$26=\frac{16+2P}{6}$$

$$16+2P=26\times 6$$
$$16+2P=156$$
$$2P=156-16$$
$$2P=140$$
$$P=70$$

Now value of observations will be

3, 3, 70, 7, (70 –1), 4

= 3, 3, 70, 7, 69, 4.

Also, Mode is 3, since it occurs most.

Q11.(b) Find the median of the following data:

Class interval	5-10	10-15	15-20	20-25	25-30	30-35
Frequency	3	16	26	31	16	8

Ans.

Class	Frequency	Cumulative Frequency
5-10	3	3
10-15	16	19
15-20	26	45
20-25	31	76
25-30	16	92
30-35	8	100

Here $\quad$ n = 100

$\frac{n}{2}$th i.e. 50th value lies in class 20- 25, which is the median class.

$\lambda = 20$

$C = 45$

$f = 31$

$c = 5$

Hence,

$$\text{median} = \lambda + \frac{\frac{n}{2} - C}{f} \times c$$

$$= 20 + \frac{50 - 45}{31} \times 5$$

$$= 20.8$$